Req ID	Data Requirement	Data Type & Format	Source / Origin	Input / Trigger	Processing & Business Rules	Output Information	Destination	Storage / Entity	Security	Frequency & Volume
IR-ORD-01	User Cart Items & Quantities	JSON Array [{item_id, qty}]	Web / Mobile Frontend	Checkout Button Click	Validate item stock, check pricing active status, apply promo code discounts.	Validated Order Breakdown (Subtotal, Tax, Shipping)	Order Review Screen	cart_session (Redis Cache)	Internal	High / ~100k requests/day
IR-ORD-02	Customer Shipping & Billing Address	Struct / Object	Customer Input Form	Address Submission	Validate address postal format, estimate shipping charges via carrier API.	Validated Address & Shipping Options	Checkout Step 2 UI	user_addresses (PostgreSQL)	PII (Confidential)	Real-time / ~20k requests/day
IR-ORD-03	Payment Token & Authorization	String (Tokenized Hash)	Payment Gateway SDK	"Place Order" Action	Transmit token securely to Stripe API; verify transaction response code.	Payment Confirmation / Decline Reason	Order Confirmation Screen & Email Service	transactions table	High Security (PCI-DSS)	Real-time / ~15k requests/day
IR-ORD-04	Order Receipt Generation	PDF Document / HTML Payload	Order Processing Service	Payment Success Event	Aggregate order items, tax summary, customer details into structured PDF template.	Downloadable Receipt URL & Email Attachment	Customer Email & User Dashboard	receipts (AWS S3 Storage)	Confidential	Batch/Async / ~15k documents/day